

# SIMATS ENGINEERING

## ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

NAME: P. Moshiya

REG NO: 192572142

---

### COURSE OUTCOME COVERED

**CO4:** *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

*Assessment Tool Weightage: 10%*

---

**Duration : 1 hour**

**QUESTIONS: 4**

**Total Marks: 40**

Annexure A

### ARTICLE REVIEW

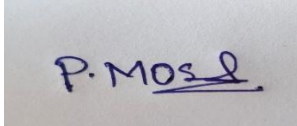
---

#### Code of Conduct:

*I P. Moshiya (192572142) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student:**



P. Mosh

## ARTICLE REVIEW

---

### **INSTRUCTIONS**

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

*Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.*

### **Task 1: Article Summary**

**(a)** Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

**(b)** Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

### **Task 2: Methodology Analysis**

**(a)** Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

**(b)** Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

### **Task 3: Results & Critical Evaluation**

**(a)** Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

**(b)** Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

**(c)** Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

### **Task 4: Reflection & Learning Outcome**

**(a)** State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

**(b)** If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

---

**RUBRICS FOR CALCULATION:**

| <b>Criteria</b>          | <b>Weightage (%)</b> | <b>Level 4 (Exemplary)</b>                                                                      | <b>Level 3 (Proficient)</b>                          | <b>Level 2 (Developing)</b>              | <b>Level 1 (Beginning)</b>               |
|--------------------------|----------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------|------------------------------------------|
| Understanding of Article | 20%                  | Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments | Shows good understanding with minor gaps             | Partial understanding; misses key ideas  | Lacks understanding of the article       |
| Summary                  | 20%                  | Provides concise and accurate summary highlighting all key points                               | Covers most key points with minor omissions          | Incomplete or partially accurate summary | Poor or incorrect summary                |
| Critical Analysis        | 25%                  | Provides insightful critique with strong reasoning and evaluation                               | Provides reasonable analysis with some justification | Superficial analysis; limited evaluation | No meaningful analysis                   |
| Use of Evidence          | 15%                  | Effectively uses examples, data, or references to support review                                | Uses some supporting evidence with minor gaps        | Limited or weak supporting evidence      | No supporting evidence provided          |
| Organization             | 20%                  | Well-structured, clear, and coherent writing with proper language and flow                      | Generally clear with minor issues                    | Poor organization; lacks clarity         | Disorganized and difficult to understand |

**SIMATS ENGINEERING**  
**SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES**  
**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

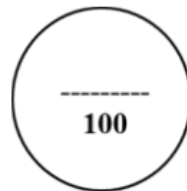
**CO4 – ASSESSMENT TOOL 3**

**Article Review**

**COURSE NAME:** Artificial Intelligence  
**QUESTIONS:** 4      **Duration:** 1 Hour

**COURSE CODE:** CSA17  
**Total Marks:** 40

| Criteria                 | Weightage | Task 1<br>(10) | Task 2<br>(10) | Task 3<br>(10) | Task 4<br>(10) | Total<br>Marks (40) |
|--------------------------|-----------|----------------|----------------|----------------|----------------|---------------------|
| Understanding of Article | 20% (2)   | /2             | /2             | /2             | /2             | /8                  |
| Summary                  | 20% (2)   | /2             | /2             | /2             | /2             | /8                  |
| Critical Analysis        | 25% (2.5) | /2.5           | /2.5           | /2.5           | /2.5           | /10                 |
| Use of Evidence          | 15% (1.5) | /1.5           | /1.5           | /1.5           | /1.5           | /6                  |
| Organization             | 20% (2)   | /2             | /2             | /2             | /2             | /8                  |
| Total per Question       | 10 Marks  | /10            | /10            | /10            | /10            | /40                 |



**TASK 1: ARTICLE SUMMARY**

**(a) Full Citation, Domain and ML Problem**

The selected research article is:

Awan, S. E., Bennamoun, M., Soheli, F., Sanfilippo, F., & Dwivedi, G. (2022). “A Reinforcement Learning-Based Approach for Imputing Missing Data.” Neural Computing and Applications, 34, 9701–9716. DOI: 10.1007/s00521-022-06958-3.

The article belongs to the domain of Machine Learning and Data Pre-processing. The main application area is missing data imputation. Missing values are frequently found in healthcare, finance, IoT, scientific, and business datasets. If missing values are not handled properly, they can negatively affect the performance of machine-learning models.

The specific ML problem addressed in this research is the estimation of missing numerical values. The authors propose a Reinforcement Learning-based approach using Q-learning to estimate missing values more effectively than simple methods such as mean, median, and most-frequent-value imputation.

## **(b) Key Objective and Research Gap**

The main objective of the research is to develop a Reinforcement Learning-based method for imputing missing numerical data. Traditional methods such as mean and median imputation usually replace all missing values in a feature with the same value. This can reduce the natural variation of the dataset and may affect the accuracy of subsequent machine-learning models. Other machine-learning approaches such as k-Nearest Neighbours and Random Forest can provide better estimates but may require more computational resources.

The research gap identified by the authors is the lack of an effective univariate imputation method that can estimate different values for different missing entries while maintaining the variation of the original data. To address this problem, the authors proposed a Q-learning-based approach. The RL agent learns through states, actions, and rewards to improve the initial estimate of each missing value. The proposed method was evaluated using eight datasets from the UCI Machine Learning Repository and compared with several existing imputation techniques. The results showed that the proposed method achieved competitive performance and outperformed several existing approaches.

## **TASK 2: METHODOLOGY ANALYSIS**

### **(a) ML Technique Used**

The main technique used in the article is Reinforcement Learning, specifically the Q-learning algorithm.

Q-learning is a model-free Reinforcement Learning algorithm in which an agent learns the best action by interacting with an environment and receiving rewards.

In this research, the RL agent is used to improve the estimated value of a missing data entry.

The basic process is:

1. The missing value is initially estimated using the mean of the feature.
2. The current estimation is treated as the state of the system.
3. The RL agent selects an action to modify the estimated value.
4. A reward is calculated based on the improvement in the estimation.
5. The Q-table is updated.
6. The agent gradually learns the best policy for estimating missing values.
7. The learned policy is used to generate the final imputed value.

Thus, the algorithm uses the concept of trial-and-error learning to improve missing-value estimation.

## Dataset Used

The researchers evaluated the proposed method using eight datasets obtained from the UCI Machine Learning Repository.

The datasets included:

| Dataset                | Instances | Attributes |
|------------------------|-----------|------------|
| Breast Cancer          | 569       | 30         |
| Vehicle                | 946       | 18         |
| Travel                 | 980       | 10         |
| Spambase               | 4,601     | 57         |
| Parkinson Disease      | 5,875     | 19         |
| Letter Recognition     | 20,000    | 17         |
| Default Credit Card    | 30,000    | 24         |
| Online News Popularity | 39,644    | 61         |

The experiments introduced missing values at different levels, including 5%, 10%, 15%, and 20%.

The data were divided into 70% training data and 30% testing data. Missing values were artificially introduced into the datasets to evaluate how accurately the proposed method could recover the original values.

The performance was measured mainly using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). Lower values of MAE and RMSE indicate better imputation performance.

### **(b) Mapping to CO 4**

The article is directly related to the Reinforcement Learning topic in CO 4.

The major CO 4 concepts demonstrated in the paper are:

- Reinforcement Learning
- Q-learning
- State

- Action
- Reward
- Policy
- Trial-and-error learning

### **Methodological Strength**

One important strength is that the proposed approach can estimate different values for different missing entries, instead of replacing all missing values in a feature with one fixed mean or median value. This helps preserve the variation in the dataset.

### **Methodological Limitation**

One important limitation is that the proposed approach mainly focuses on numerical data. It does not directly handle categorical variables. Another limitation is that the experiments mainly use artificially generated missing values, which may not completely represent missing values found in real-world datasets.

## **TASK 3: RESULTS AND CRITICAL EVALUATION**

### **(a) Key Results**

The authors compared their proposed Reinforcement Learning approach with several existing techniques, including:

- Mean imputation
- Median imputation
- Most-Frequent imputation
- k-Nearest Neighbours
- Random Forest
- MICE
- GAIN
- CGAIN

The proposed Reinforcement Learning approach performed strongly across the experiments. According to the authors, it outperformed the other approaches on six of the eight datasets and ranked among the top three methods on the remaining two datasets.

**Example:** Spambase Dataset with 5% Missing Data

| <b>Method</b>      | <b>MAE</b>    | <b>RMSE</b>   |
|--------------------|---------------|---------------|
| Mean               | 0.0271        | 0.0544        |
| Median             | 0.0201        | 0.0591        |
| Most Frequent      | 0.0208        | 0.0615        |
| kNN                | 0.0309        | 0.0719        |
| Random Forest      | 0.0309        | 0.0696        |
| MICE               | 0.0286        | 0.0588        |
| GAIN               | 0.0501        | 0.0723        |
| CGAIN              | 0.0447        | 0.0611        |
| <b>Proposed RL</b> | <b>0.0183</b> | <b>0.0485</b> |

The proposed RL method obtained the lowest MAE and RMSE in this experiment.

Spambase Dataset with 20% Missing Data

| <b>Method</b> | <b>MAE</b> | <b>RMSE</b> |
|---------------|------------|-------------|
| Mean          | 0.0278     | 0.0593      |
| Median        | 0.0210     | 0.0635      |
| Most Frequent | 0.0217     | 0.0667      |
| kNN           | 0.0319     | 0.0750      |
| Random Forest | 0.0321     | 0.0715      |
| MICE          | 0.0290     | 0.0620      |
| GAIN          | 0.0595     | 0.0764      |
| CGAIN         | 0.0430     | 0.0601      |

| Method | MAE | RMSE |
|--------|-----|------|
|--------|-----|------|

|             |        |        |
|-------------|--------|--------|
| Proposed RL | 0.0198 | 0.0527 |
|-------------|--------|--------|

The RL-based method again produced the lowest MAE and RMSE among the compared methods.

The authors also repeated experiments multiple times to check the stability of the proposed method. The results showed relatively small variation in the reported MAE and RMSE values.

## **(b) Critical Evaluation**

### **Are the Reported Metrics Realistic?**

The reported MAE and RMSE values are reasonable for the experimental setup because the authors know the original values before artificially removing them. Therefore, the estimated values can be compared directly with the actual values.

However, these results should be interpreted carefully because artificially generated missing values may not behave exactly like naturally occurring missing values.

### **Potential Biases**

There are some potential limitations and biases in the evaluation.

#### **1. Artificial Missing Values**

The missing values were artificially introduced into complete datasets. Real-world missing data may occur for specific reasons and may not be randomly distributed.

#### **2. Dataset Bias**

The experiments use public UCI datasets. The results may not automatically generalize to large industrial datasets.

#### **3. Numerical Data Limitation**

The proposed approach primarily focuses on numerical variables, limiting its direct application to datasets containing categorical variables.

#### **4. Evaluation Protocol**

The method could be evaluated further using additional validation strategies such as k-fold cross-validation and independent external datasets.

### **Additional Experiments Required**

The research could be strengthened by:

1. Testing the method on real-world datasets containing naturally missing values.
2. Testing Missing At Random and Missing Not At Random conditions.
3. Including categorical and mixed-type datasets.
4. Comparing with recent deep-learning-based imputation methods.
5. Testing the algorithm on very large industrial datasets.
6. Performing statistical significance tests.
7. Conducting an ablation study to determine which RL parameters contribute most to performance.

### **(c) Real-World Applicability**

The proposed method can be useful in many real-world applications where datasets contain missing numerical values.

#### **Possible Applications**

**Healthcare:** Missing patient measurements such as laboratory values or physiological measurements can be estimated before applying predictive models.

**Finance:** Missing financial and customer information can be handled before credit-risk analysis.

**IoT:** Missing sensor readings can be estimated to improve monitoring and prediction.

**Industry:** Missing machine and equipment measurements can be imputed for predictive maintenance.

**Business Analytics:** Missing customer or transaction information can be estimated before performing data analysis.

#### **Challenges in Industry**

The main challenges include:

- Availability of high-quality training data
- Large-scale computational requirements
- Different types of missing-data patterns
- Handling categorical and mixed data
- Data privacy and security
- Reliability of estimated values

- Integration with existing industrial ML systems

For critical applications such as healthcare and finance, incorrect imputation could affect decision-making. Therefore, uncertainty estimation and human supervision may also be required.

## **TASK 4: REFLECTION AND LEARNING OUTCOME**

### **(a) Three Key Insights**

#### **Insight 1: Reinforcement Learning Has Applications Beyond Games**

The article helped me understand that Reinforcement Learning is not limited to games, robotics, or autonomous systems. It can also be applied to data-preprocessing tasks such as missing-value imputation.

**CO 4 Concept:** Reinforcement Learning and real-world applications.

#### **Insight 2: Q-Learning Learns Through Rewards**

I learned how an RL agent can improve its decisions through trial and error. In this article, the agent selects actions to modify the estimated missing value and receives a reward based on the quality of the estimation.

**CO 4 Concept:** Q-learning, state, action, reward, and policy.

#### **Insight 3: Performance Metrics Depend on the ML Problem**

The article helped me understand that the evaluation metric must be selected according to the type of problem. Since missing-value imputation is a numerical prediction problem, MAE and RMSE are more appropriate than classification metrics such as accuracy.

**CO 4 Concept:** Machine-learning model evaluation.

### **(b) Proposed Extension**

I would extend this research by developing a Reinforcement Learning-based imputation method for mixed numerical and categorical datasets.

The current approach mainly focuses on numerical variables. In real-world applications, datasets often contain both numerical and categorical attributes. For example, a healthcare dataset may contain age and blood pressure as numerical variables, while gender, disease type, and treatment category may be categorical variables.

The proposed experiment would modify the Q-learning action space so that the agent can select numerical adjustments for numerical features and category-selection actions for categorical features.

The experiment could be conducted using a real-world healthcare or financial dataset. Missing values could be introduced at different levels, such as 5%, 10%, 15%, and 20%. MAE and RMSE could be used for numerical variables, while accuracy and F1-score could be used for categorical variables.

This extension is feasible because the existing Q-learning framework can be modified to support different types of actions. It would make the proposed method more practical for real-world datasets and increase its applicability in healthcare, finance, IoT, and business analytics.